Assignment 1.1- Linux Installation and Administration

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Phase 1: Linux Installation

Ubuntu Server 24.04 LTS was set up on Oracle Virtual box prompting Full disk encryption (FDE) passphrase during system boot. This screenshot confirms that Full Disk Encryption was enabled, requiring for the disk to be unlocked before operating system can load. Thereby ensuring confidentiality of data at rest.

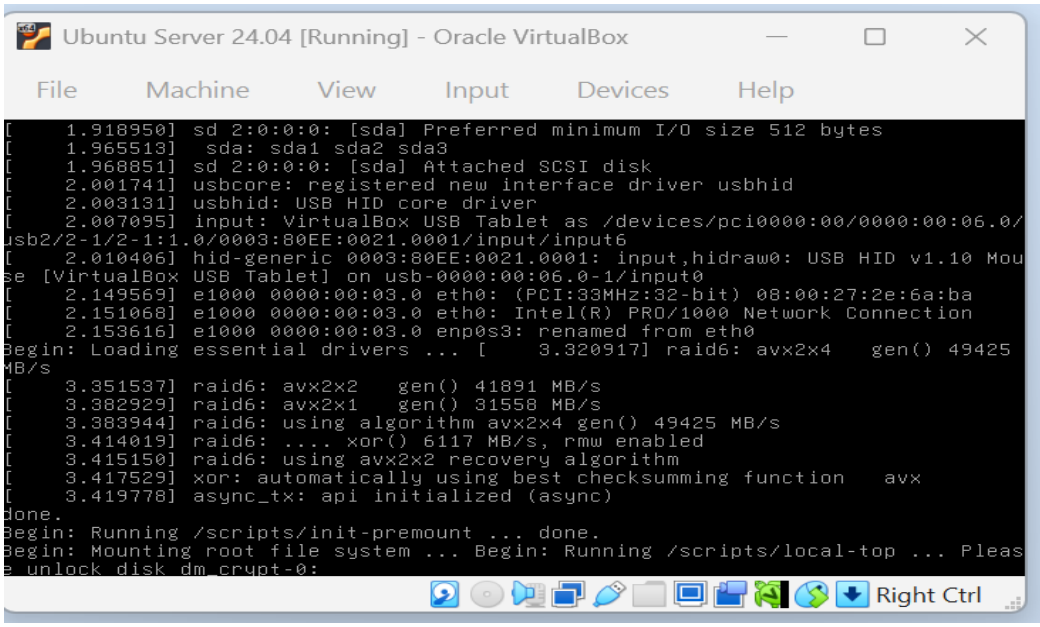
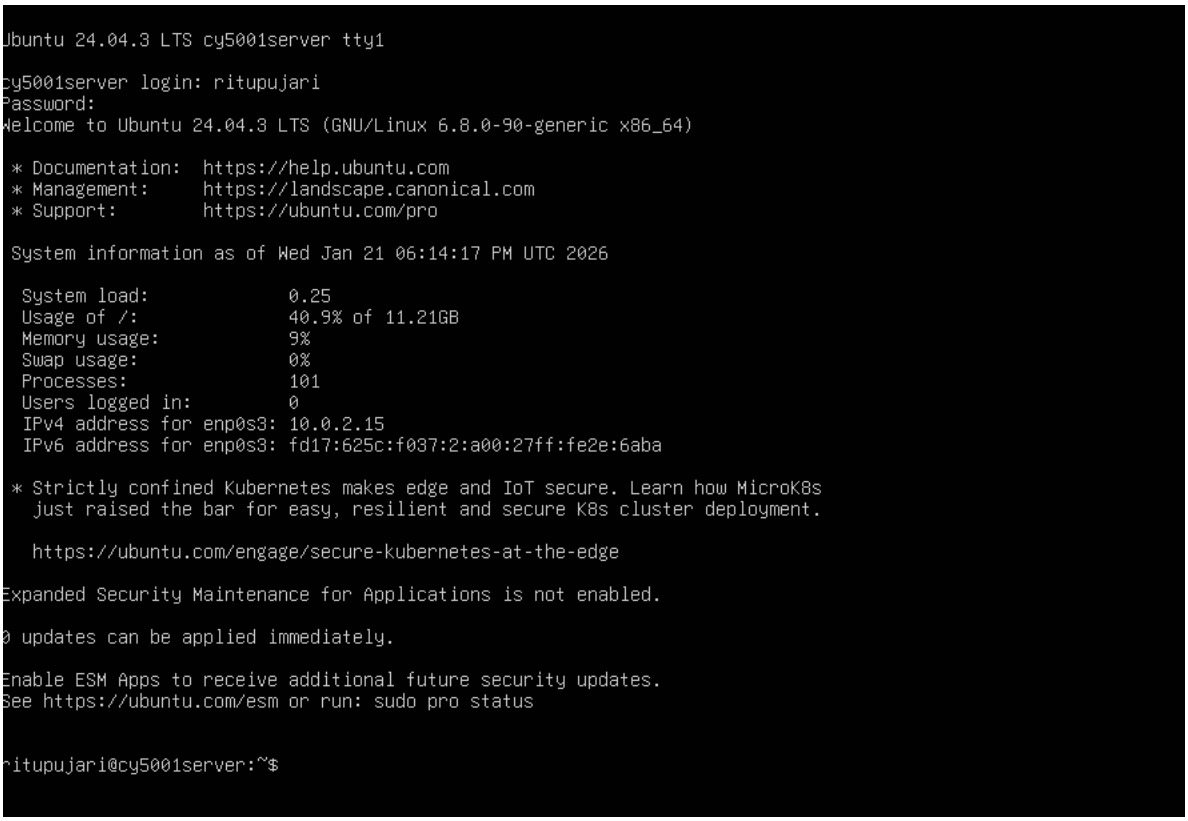


Figure 1.1: Full Disk Encryption Password Prompt During First Boot



```
Ubuntu 24.04.3 LTS cy5001server tty1

cy5001server login: ritupujari
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan 21 06:14:17 PM UTC 2026

  System load:             0.25
  Usage of /:              40.9% of 11.21GB
  Memory usage:            9%
  Swap usage:              0%
  Processes:               101
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe2e:6aba

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


ritupujari@cy5001server:~$
```

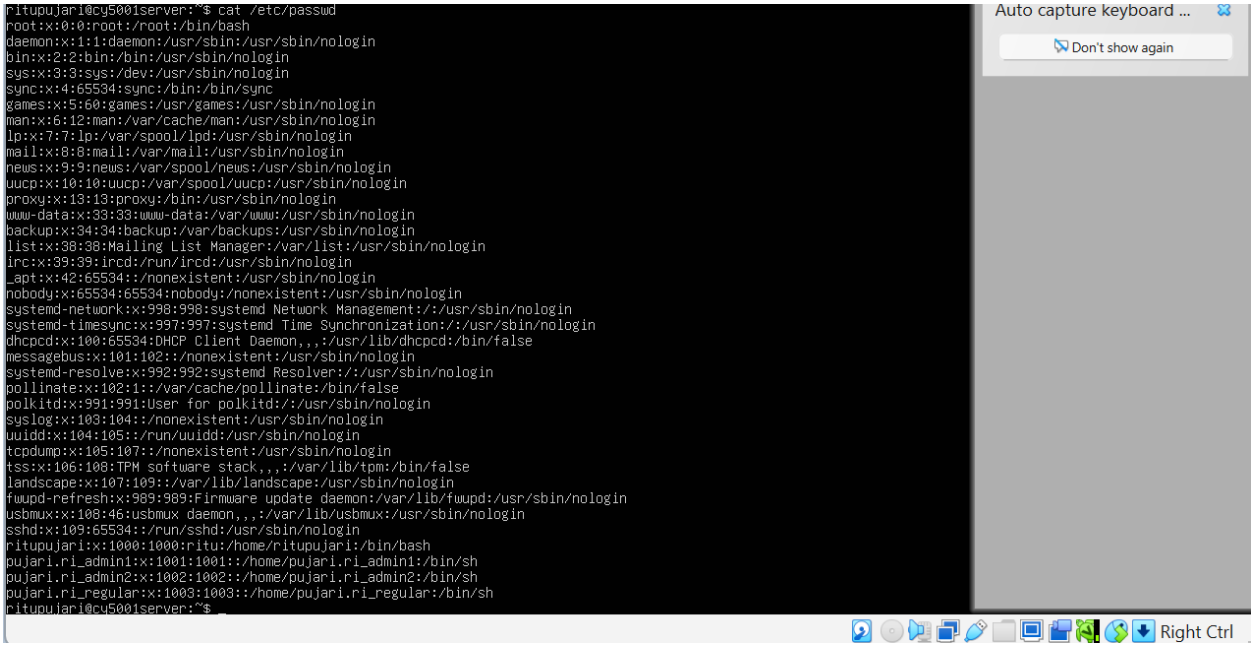
Figure 1.2: Ubuntu Server Booted and Logged in as ritupujari

The screenshot displays a successful login into the Ubuntu Server System using the user account created during the installation.

Phase 2: Administrative Tasks

Task 2- Create Users

Figure 2.1 shows the contents of /etc/passwd after creating the required user accounts – *pujari.ri_admin1, pujari.ri_admin2, pujari.ri_regular* and they are visible in the file, confirmimg the new users were successfully created.



```
ritupujari@cy5001server:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
ritupujari:x:1000:1000:ritu:/home/ritupujari:/bin/bash
pujari.ri_admin1:x:1001:1001::/home/pujari.ri_admin1:/bin/sh
pujari.ri_admin2:x:1002:1002::/home/pujari.ri_admin2:/bin/sh
pujari.ri_regular:x:1003:1003::/home/pujari.ri_regular:/bin/sh
ritupujari@cy5001server:~$
```


*Figure 2.1: Output of /**etc/passwd** showing newly created users*

Task 3 - Create a new group of users

A new group named *superusers* was created with members - pujari.ri_admin1, pujari.ri_admin2

Task 4 - Control access to a folder directory in the system

Figure 2.2 shows the permission settings for the directory *justfor_pujari.ri* where complete access of read, write and execute are granted to the members of superusers group and limits the regular user *pujari.ri_regular* to read-only access.

```
$ su - pujari.ri_admin1
Password:
$ mkdir ~/justfor_pujari.ri
$ chown pujari.ri_admin1:superusers ~/justfor_pujari.ri
$ chmod 770 ~/justfor_pujari.ri
$ setfacl -m u:pujari.ri_regular:r ~/justfor_pujari.ri
$ ls -ld ~/justfor_pujari.ri
drwxrwx---+ 2 pujari.ri_admin1 superusers 4096 Jan 21 04:05 /home/pujari.ri_admin1/justfor_pujari.ri
$ getfacl ~/justfor_pujari.ri
getfacl: Removing leading '/' from absolute path names
# file: home/pujari.ri_admin1/justfor_pujari.ri
# owner: pujari.ri_admin1
# group: superusers
user::rwx
user:pujari.ri_regular:r--
group::rwx
mask::rwx
other::---
```

Figure 2.2: Access Control Permissions for justfor_pujari.ri

Task 5 - Switch user and create a file

Figure 2.3 illustrates the permission settings for the file *pujari.ri_file* created by the user *pujari.ri_admin1* within the directory *justfor_pujari.ri*. The file permissions indicate that the owner has read and write access, while group members and other users have read only access.

```
$ chmod 644 pujari.ri_file
$ ls -l pujari.ri_file
-rw-r--r-- 1 pujari.ri_admin1 pujari.ri_admin1 58 Jan 21 04:11 pujari.ri_file
$ whoami
pujari.ri_admin1
```

Figure 2.3: Permissions for pujari.ri_file

Task 6 - Read a different user's file.

```
ritupujari@cy5001server:~$ su - pujari.ri_regular
Password:
$ whoami
pujari.ri_regular
$ cat /home/pujari.ri_admin1/justfor_pujari.ri/pujari.ri_file
cat: /home/pujari.ri_admin1/justfor_pujari.ri/pujari.ri_file: Permission denied
```

Figure 2.4: Access Denied when pujari.ri_regular attempts to read protected pujari.ri_file

Access is denied since *pujari.ri_regular* is not a member of the superusers group and the directory *justfor_pujari.ri* does not give execute (x) permissions to the regular users. Execute permissions on a directory in Linux give users the ability to traverse that directory. Since the regular user is not a member of the group and cannot execute the directory, the system denies the user access to the file.

Task 7 - Write a simple text file

```
$ whoami
pujari.ri_regular
$ cat /home/pujari.ri_admin1/justfor_pujari.ri/pujari.ri_file
cat: /home/pujari.ri_admin1/justfor_pujari.ri/pujari.ri_file: Permission denied
$ echo "test file" >> /home/pujari.ri_admin1/justfor_pujari.ri/test.txt
-sh: 3: cannot create /home/pujari.ri_admin1/justfor_pujari.ri/test.txt: Permission denied
```

Figure 2.5: Access Denied when pujari.ri_regular attempts to write in the protected pujari.ri_file

The permission to write was denied because *pujari.ri_regular* is not a member of the superusers group, and the directory *justfor_pujari.ri* doesn't grant write (w) privileges for the regular users.

Task 8 - Try to delete a file

The delete operation fails because the *pujari.ri_regular* does not have the write (w) privilege for the directory *justfor_pujari.ri*. The delete command in Linux needs the write and execute permissions.

```
$ rm /home/pujari.ri_admin1/justfor_pujari.ri/pujari.ri_file
rm: cannot remove '/home/pujari.ri_admin1/justfor_pujari.ri/pujari.ri_file': Permission denied
```

Figure 2.6: Access Denied when pujari.ri_regular attempts to delete the pujari.ri_file

Task 9 - Check audit logs

Login operations were checked using */var/log/auth.log*. The log entries indicates that user authentication operations are audited correctly by the system.

```
$ exit
ritupujari@cy5001server:~$ sudo tail /var/log/auth.log
[sudo] password for ritupujari:
Sorry, try again.
[sudo] password for ritupujari:
2026-01-21T04:13:35.315224+00:00 cy5001server su[2498]: (to pujari.ri_regular) ritupujari on pts/0
2026-01-21T04:13:35.316810+00:00 cy5001server su[2498]: pam_unix(su-l:session): session opened for user pujari.ri_regular(uid=1003) by ritupujari(uid=1000)
2026-01-21T04:15:01.790511+00:00 cy5001server CRON[2506]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-01-21T04:15:01.811485+00:00 cy5001server CRON[2506]: pam_unix(cron:session): session closed for user root
2026-01-21T04:17:01.811411+00:00 cy5001server CRON[2513]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-01-21T04:17:01.835367+00:00 cy5001server CRON[2513]: pam_unix(cron:session): session closed for user root
2026-01-21T04:20:31.237162+00:00 cy5001server su[2498]: pam_unix(su-l:session): session closed for user pujari.ri_regular
2026-01-21T04:21:31.262349+00:00 cy5001server sudo: pam_unix(sudo:auth): authentication failure; logname=ritupujari uid=1000 euid=0 tty=/dev/pts/0 ruser=ritupuj
ari rhost=  user=ritupujari
2026-01-21T04:21:39.170306+00:00 cy5001server sudo: ritupujari : TTY=pts/0 ; PWD=/home/ritupujari ; USER=root ; COMMAND=/usr/bin/tail /var/log/auth.log
2026-01-21T04:21:39.172528+00:00 cy5001server sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ritupujari(uid=1000)
ritupujari@cy5001server:~$
```

Figure 2.7: Login Events recorded in System Logs

Task 10 - (Check known vulnerabilities) Install security updates

The security updates were checked and applied using the command *sudo apt update* followed by the command *sudo apt upgrade -y*. From the output, it can be seen that the system is fully up to date with no pending updates or upgrades. This shows that the system is protected against known vulnerabilities at the time of execution.

```
ritupujari@cy5001server:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ritupujari@cy5001server:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Figure 2.8: Operating System Security Updates

Task 11 - (Confidentiality) Install eCryptfs in Linux and create an encrypted folder directory

```
ritupujari@cy5001server:~$ mkdir ~/CY5001
ritupujari@cy5001server:~$ sudo mount -t ecryptfs ~/CY5001 ~/CY5001
Select key type to use for newly created files:
 1) passphrase
 2) tspi
Selection: 1
Passphrase:
Select cipher:
 1) aes: blocksize = 16; min keysize = 16; max keysize = 32
 2) blowfish: blocksize = 8; min keysize = 16; max keysize = 56
 3) des3_ede: blocksize = 8; min keysize = 24; max keysize = 24
 4) twofish: blocksize = 16; min keysize = 16; max keysize = 32
 5) cast6: blocksize = 16; min keysize = 16; max keysize = 32
 6) cast5: blocksize = 8; min keysize = 5; max keysize = 16
Selection [aes]: 1
Select key bytes:
 1) 16
 2) 32
 3) 24
Selection [16]: 2
Enable plaintext passthrough (y/n) [n]: n
Enable filename encryption (y/n) [n]: n
Attempting to mount with the following options:
  ecryptfs_unlink_sigs
  ecryptfs_key_bytes=32
  ecryptfs_cipher=aes
  ecryptfs_sig=b08f22529e5dc29f
WARNING: Based on the contents of [/root/.ecryptfs/sig-cache.txt],
it looks like you have never mounted with this key
before. This could mean that you have typed your
passphrase wrong.

Would you like to proceed with the mount (yes/no)? : y
Would you like to proceed with the mount (yes/no)? : y
Would you like to proceed with the mount (yes/no)? : yes
Would you like to append sig [b08f22529e5dc29f] to
[/root/.ecryptfs/sig-cache.txt]
in order to avoid this warning in the future (yes/no)? : yes
Successfully appended new sig to user sig cache file
Mounted eCryptfs
ritupujari@cy5001server:~$
```

Right Ctrl

Task 12 - (Confidentiality) Encrypt a file stored into the CY5001 directory

The file contained within the CY5001 directory can be read while the directory is mounted.

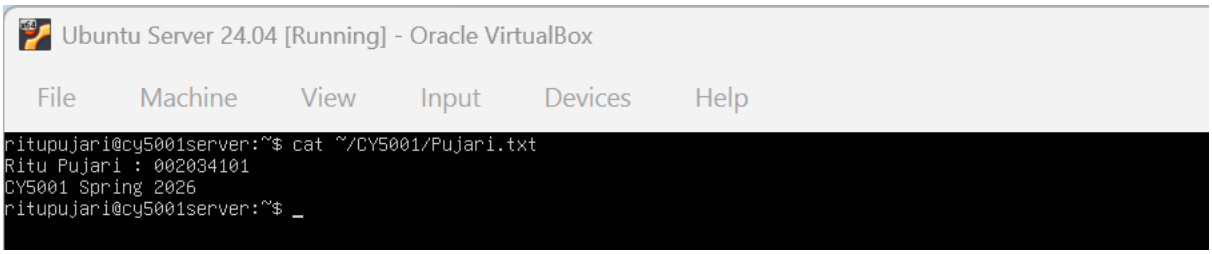


Figure 2.9: Viewing file content while directory is mounted

Task 13 - Unmount your encrypted directory

As shown below *(Fig 2.10)* after the CY5001 directory is unmounted, the files are no longer readable, thus proving that the files are encrypted and not accessible unless they are remounted using the right password. Even for superusers *(Fig 2.11)*, the encrypted file cannot be read after unmounting. This shows that encryption is independent of the privileges of the user at the filesystem level.

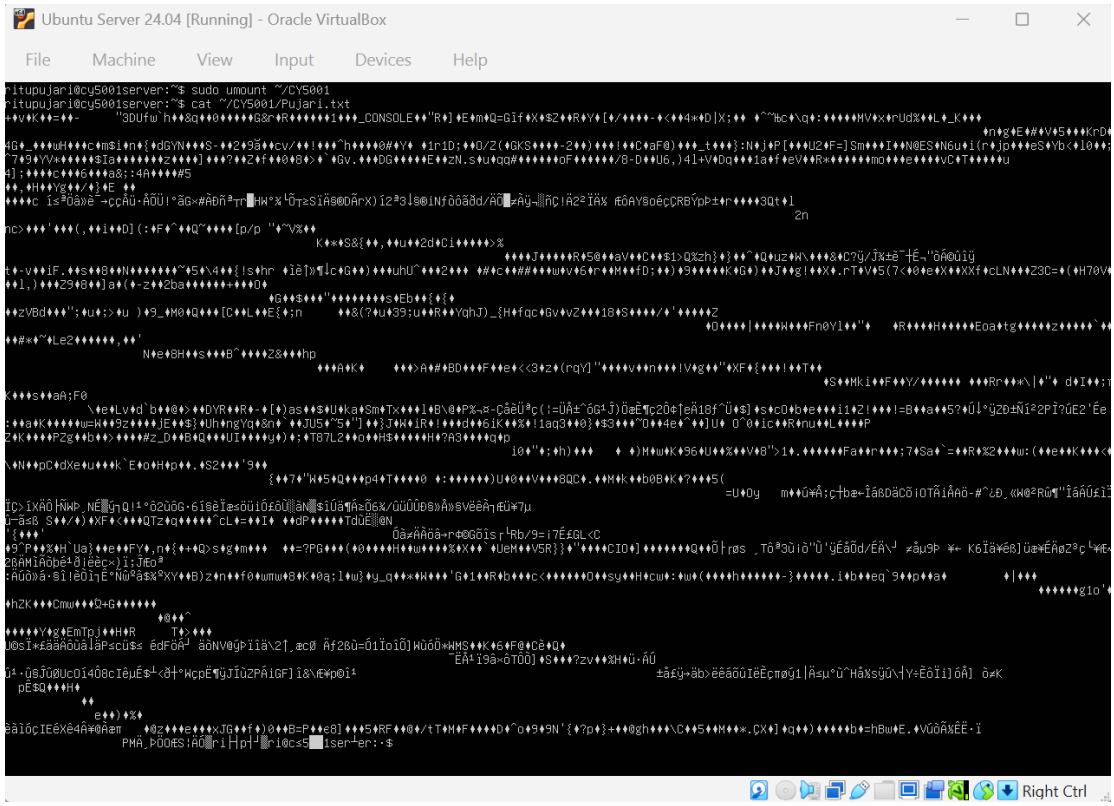


Figure 2.10: Unmounting the directory and viewing the file content

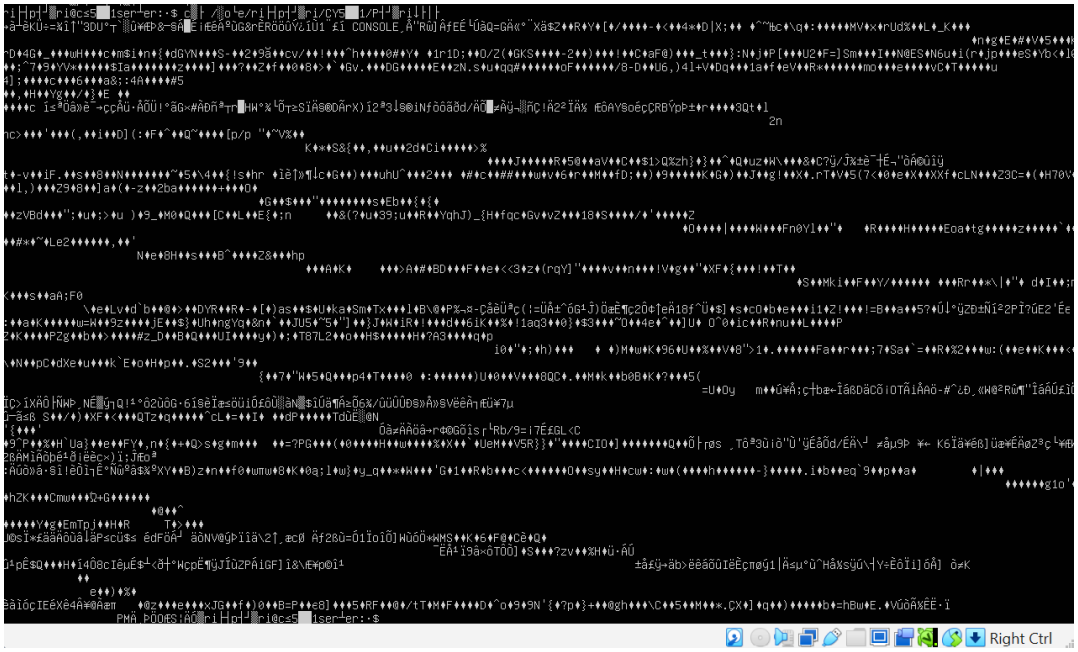


Figure 2.11: Viewing file content with superuser privilege